Future-proofing the IT career path OF Sri Lanka

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Project Proposal Report

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

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Declaration of the Candidate & Supervisor

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

	Name	Signature	Date
Supervisor	Ms.Sandeepa Dushanthi kuruppu	13/03/2026	

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ABSTRACT

Preparing for technical interviews remains a major challenge for Sri Lankan IT undergraduates because most existing preparation methods rely on static question lists, generic mock interviews, and delayed human feedback. Within the broader group research project on skill forecasting and career pathway optimization, this component focuses on building an intelligent interview simulation module that dynamically generates role-relevant IT interview questions and analyzes the candidate's facial emotions and engagement patterns during responses. The proposed solution consists of a web-based application with webcam access, real-time interview question delivery, large language model based question generation, and computer-vision-based emotion tracking. The system will use pre-trained facial analysis models to detect visible emotional states and response-related engagement indicators, log temporal changes during each answer, and provide immediate visual feedback through dashboards. The output will help users understand performance trends such as confidence fluctuations, hesitation points, and emotional consistency across different questions. Evaluation will be conducted through mock interview sessions with volunteer participants and compared against human evaluator observations to assess usefulness, relevance, and reliability. The expected outcome of this component is a practical, localized interview preparation tool that goes beyond conventional mock interview platforms by combining dynamic questioning with emotion-aware analytics for personalized feedback. This contributes directly to the group's integrated AI-powered career support ecosystem for Sri Lankan IT undergraduates.

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1. INTRODUCTION

1.1.BACKGROUND & LITERATURE SURVEY

The transition from university education to the professional IT industry in Sri Lanka is increasingly difficult due to skill mismatches, changing job market expectations, and the lack of personalized career guidance tools for undergraduates. The overall group project addresses this challenge by proposing an AI-powered ecosystem that integrates skill forecasting, semantic career pathway prediction, CV optimization, and interview simulation in a single platform. Within that ecosystem, the interview simulation component plays an important role by helping students prepare for interviews using dynamic question delivery and performance feedback.

According to the topic assessment form, the broader solution already identifies interview simulation as a key output of the integrated dashboard, where large language model APIs and computer vision techniques are combined to provide interview Q&A, emotion feedback, and performance metrics. The system-level design also shows that user-generated webcam feeds are an input to the AI processing core, while interview simulator interfaces appear as one of the main outputs to end users. This confirms that the proposed module is not isolated, but a core sub-module of the full research system.

Prior research shows that automated interview assessment has long benefited from multimodal analysis. Earlier studies demonstrated that interview performance can be assessed using a combination of facial expressions, language features, and prosodic cues, highlighting that interview quality depends not only on what a candidate says, but also on how the response is delivered. Large-scale video interview studies further showed that facial expressions, prosody, and natural language features can be used together for interview-related evaluation and judgement [1], [2].

More recent studies have explored AI-powered mock interview systems that provide automated practice and feedback. Generative-language-model-based research has shown that prompt-driven AI evaluation can reduce manual grading effort in mock interview settings [3]. Recent system-oriented studies such as AI-PrepMate and AI-Based Mock Interview System with Emotion Detection and Real-Time Feedback demonstrate the practical feasibility of building platforms that support AI-assisted mock interviews, response evaluation, emotion-aware analysis, and performance reporting [4], [5]. In addition, multimodal mock interview research has increasingly combined facial expression analysis, speech emotion recognition, and natural language processing to provide more holistic candidate feedback [6].

Despite these advances, many existing mock interview systems still provide predefined questions, generic scoring, or isolated analysis modules. Few systems combine dynamic question generation, webcam-based behavioral analytics, and clear performance visualization within a single platform designed specifically for undergraduate interview preparation. This creates a strong need for an adaptive and localized interview-preparation module that can offer more meaningful feedback to Sri Lankan IT undergraduates.

Sources for this literature synthesis

1.2. RESEARCH GAP

Although AI-based mock interview platforms have improved interview preparation through automated scoring, conversational simulation, and feedback generation, several limitations still remain. Most existing systems focus mainly on question-answer evaluation or isolated emotion detection, rather than integrating both into a single unified interview-preparation platform. In many cases, these systems rely on static or predefined question banks, which limits their ability to adapt to the candidate's job role, technical domain, and skill level.

Another major limitation is the lack of real-time behavioral analysis during interview sessions. While some studies explore emotion recognition, practical systems rarely provide live webcam-based insights into candidate engagement, hesitation, and confidence. In addition, many existing platforms offer only simple textual comments or score-based feedback, with limited use of visual analytics to display emotional trends and question-wise performance.

A further gap is the lack of localized intelligent interview-preparation tools designed specifically for Sri Lankan IT undergraduates. Most current systems are developed for general or global contexts and do not sufficiently reflect regional career expectations, technical requirements, and undergraduate employability needs.

Therefore, this research addresses the identified gap by proposing an emotion-aware interview simulation system that integrates dynamic question generation, webcam-based behavioral analytics, and interactive performance visualization in a single platform. By combining conversational AI with computer vision in a localized academic context, the proposed study offers a more comprehensive and practical approach to improving interview readiness among Sri Lankan IT undergraduates.

Research Paper	AI Question Generation	Emotion Detection	Real-time Behavior Analysis	Analytics Dashboard	Localized System	Key Limitation
AI-Based Mock Interview System (IJSRSET, 2026)	✓	✕	✕	Basic report	✕	Uses AI for interview simulation but lacks behavioral analysis and emotion tracking.
AI-Based Mock Interview System with Emotion Detection (IJRASET, 2025)	✕	✓	Partial	✕	✕	Detects emotions but does not generate dynamic interview questions.
AI-Powered Mock Interview Platform with NLP & Speech Analysis (IRJAEH, 2025)	✓	✕	Partial	Partial	✕	Focuses on NLP evaluation and resume-based questions but lacks visual emotion analysis.
AI-Based Interview Evaluator – Emotion & Confidence Classifier (IARJSET)	✕	✓	✓	✕	✕	Evaluates facial emotion and confidence but lacks question generation and analytics visualization.
AI-Driven Mock Interview Platform (IJARCCE)	Partial	Partial	✓	✕	✕	Uses AI for behavioral evaluation but lacks integrated multimodal analytics
Multimodal AI Mock Interview System (IJSRED, 2025)	Partial	✓	✓	Partial	✕	Integrates multiple modalities but lacks interactive dashboards and localization.
AI Mock Interview Platform with Generative AI and Computer Vision (ICSSAS 2025)	✓	✓	✓	Partial	✕	Combines generative AI and CV but lacks personalized performance analytics dashboards.
Automated Online Mock Interview System with ML and Emotion Analysis (ISJEM, 2026)	✓	✓	✓	Basic	✕	Provides automated evaluation but limited visualization and system personalization
Future-proofing the IT career path OF Sri Lanka	✓	✓	✓	✓	✓	Integrates all modules in a single platform

Table 1:comparison table

1.3.RESEARCH PROBLEM

Sri Lankan IT undergraduates often prepare for interviews using generic materials and receive subjective or delayed feedback. As a result, they may know technical answers but still struggle with confidence, consistency, and communication during interviews. The key problem addressed by this component is the lack of an intelligent interview simulation system that can ask role-relevant questions in real time, observe the candidate's visible emotional and engagement patterns through webcam data, and provide immediate, actionable feedback for improvement. Solving this problem would support more effective interview preparation and strengthen the broader goal of improving graduate

2. OBJECTIVES

2.1. MAIN OBJECTIVES

To develop an emotion-aware AI interview simulation system for Sri Lankan IT undergraduates that dynamically generates interview questions, analyzes webcam-based emotional and engagement patterns during responses, and provides immediate personalized performance feedback through an interactive dashboard.

2.2. SPECIFIC OBJECTIVES

To design and implement a web application with webcam access and interview question display for mock interview sessions.

To integrate a large language model API to generate role-specific and difficulty-aware IT interview questions dynamically.

To detect and track the candidate's facial emotions and visible engagement indicators using pre-trained computer vision models such as MediaPipe and FER-based classifiers.

To log temporal emotion changes and response-level session data for each interview question.

To generate dashboards that visualize emotional trends, question-wise performance, and improvement suggestions.

To validate the usefulness and reliability of the proposed system through mock interview sessions and comparison with human evaluator feedback

3. METHODOLOGY

3.1. INTRODUCTION TO METHODOLOGY

This component will be developed as a web-based mock interview platform that combines frontend interaction, LLM-powered question generation, webcam-based facial analytics, and dashboard-based reporting. The methodology is designed to align with the group project's system perspective, where large language models and computer vision are used together to support interview simulation and performance analysis.

3.2. REQUIREMENT GATHERING

Requirements will be gathered from undergraduate students, academic supervisors, and where possible, industry professionals familiar with IT interview expectations. Key functional requirements include real-time question presentation, webcam-based session capture, interview logging, emotion analysis, and dashboard-based feedback. Non-functional requirements include low response delay, usability, privacy protection, and reliable operation on commonly available student devices. Requirement gathering will ensure that the component remains aligned with the needs of Sri Lankan IT undergraduates and the larger integrated platform.

3.3. SYSTEM OVERVIEW

The proposed system will begin with the user selecting an interview role such as software engineer, QA engineer, data analyst, or intern-level IT position. The frontend application will then request interview questions from the backend, where Gemini API prompts will generate role-relevant and progressive questions. During each answer, the webcam stream will be captured and processed to detect face presence, landmark movement, and emotion-related indicators. The system will store question-level and timestamped session logs, then generate summaries and dashboard visualizations after the interview. Final outputs will include detected emotional patterns, performance trends, and recommendations for future practice. This approach is consistent with the

group-level system design, which already includes webcam feeds as inputs and interview simulator interfaces with Q&A, emotion feedback, and metrics as outputs.

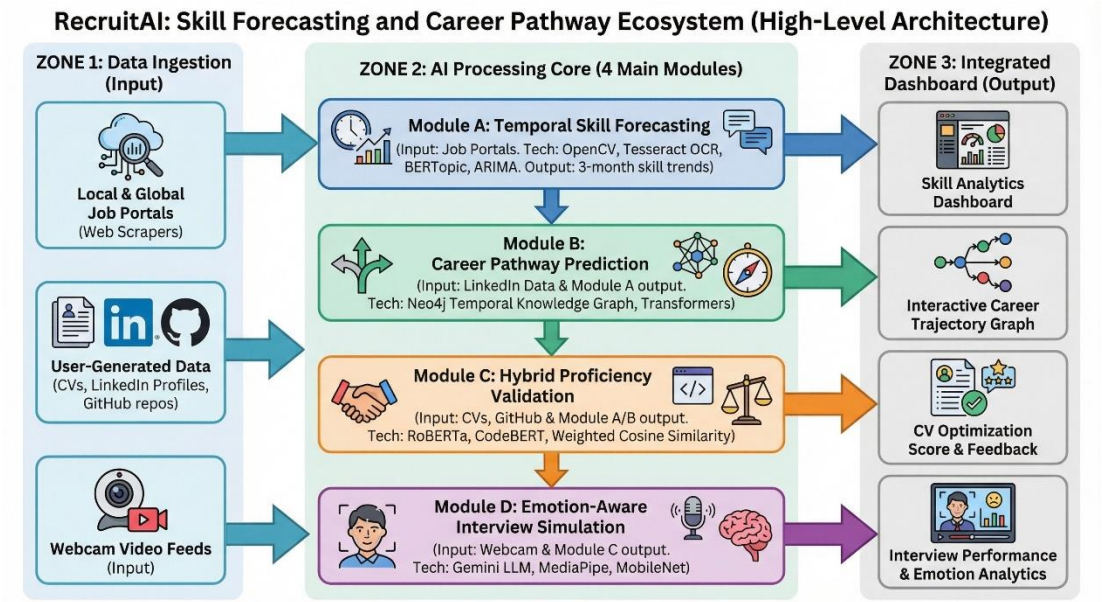


Figure 1: System Overview

3.4. COMPONENT OVERVIEW

The interview simulation module will consist of five major stages. First, the system will capture candidate details and preferred interview role. Second, Gemini API will generate a sequence of technical and behavioral interview questions. Third, while the candidate answers, the webcam feed will be analyzed using pre-trained facial models to identify visible emotional states and engagement-related cues. Fourth, the system will compute response-level analytics such as emotional consistency, hesitation-related fluctuations, and overall interview session behavior. Finally, the results will be displayed through an interactive dashboard showing timelines, per-question feedback, and practical improvement suggestions. The component directly reflects the tasks and novelty stated in your assigned sub-objective.

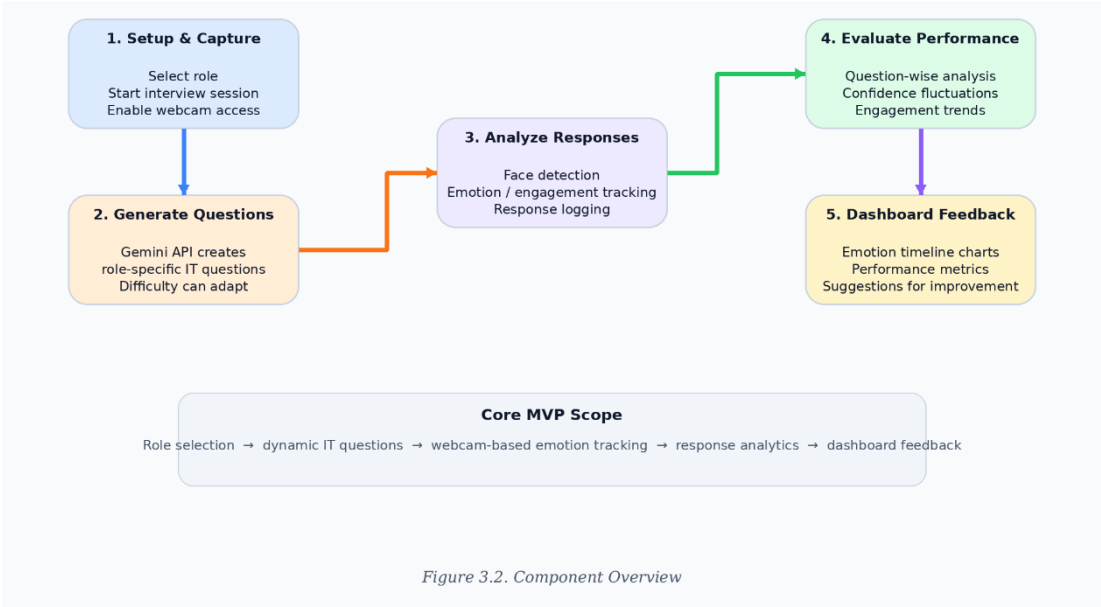


Figure 3.2. Component Overview

3.5. DATA COLLECTION AND PREPROCESSING

This component does not necessarily require large-scale model training from scratch. Instead, data collection will focus on two areas: interview content data and validation session data. Interview content data will include public IT interview question sources, job role descriptions, and prompt templates for role-specific question generation. Validation session data will be collected from voluntary mock interview participants who provide informed consent for webcam recording and analysis. Captured video frames will be sampled at fixed intervals, resized, cleaned, and processed to remove low-quality frames. Metadata such as timestamps, question IDs, answer duration, and detected emotion labels will be stored for analysis. Since the overall project also mentions consent-based user interview simulations and ethically collected user data, this approach remains aligned with the group's data practices.

3.6. MODEL DEVELOPMENT AND TRAINING

The system will use pre-trained computer vision models rather than training a new facial emotion model from the beginning, which is more practical for a one-year undergraduate component. MediaPipe can be used for face detection and landmark tracking, while an FER-based classifier can estimate visible emotion

categories such as neutral, happy, surprise, fear, sadness, or frustration depending on the selected model. Gemini API will be integrated to generate interview questions dynamically using prompts that vary by role, domain, and difficulty level. A rule-based or weighted scoring mechanism can then be used to combine detected facial indicators, session behavior, and question flow into an engagement-oriented performance summary. This approach matches the tools and technologies already mentioned in the topic assessment document, including React, Python, Gemini API, MediaPipe, and dashboard-based outputs.

3.7. SYSTEM INTEGRATION AND MOBILE APP DEVELOPMENT

The frontend can be implemented using React with webcam support through Webcam.js or native browser media APIs. The backend can be implemented using Python with FastAPI or Flask for API communication, session handling, and model orchestration. Gemini API will generate questions and optionally produce feedback summaries. Emotion analysis results and session logs will be stored in a database such as PostgreSQL or MongoDB. The dashboard will visualize question-wise performance, emotion timelines, total session summary, and recommendations using charts and tables. Because the group-level architecture already emphasizes an integrated dashboard and interview simulator interface, this part will also contribute directly to the unified platform vision.

3.8. TESTING AND VALIDATION

System validation will be carried out using mock interview sessions with volunteer participants. Functional testing will verify webcam access, question loading, logging, and dashboard generation. Performance testing will measure latency and session

stability. Usability testing will evaluate whether students find the feedback understandable and useful. Human evaluator comparison will be used to assess whether the generated insights correspond reasonably with observed interview behavior. The assigned tasks in the topic form specifically mention mock interview validation and human evaluator comparison, so this should be included clearly in your proposal.

3.9. PROJECT PLANNING

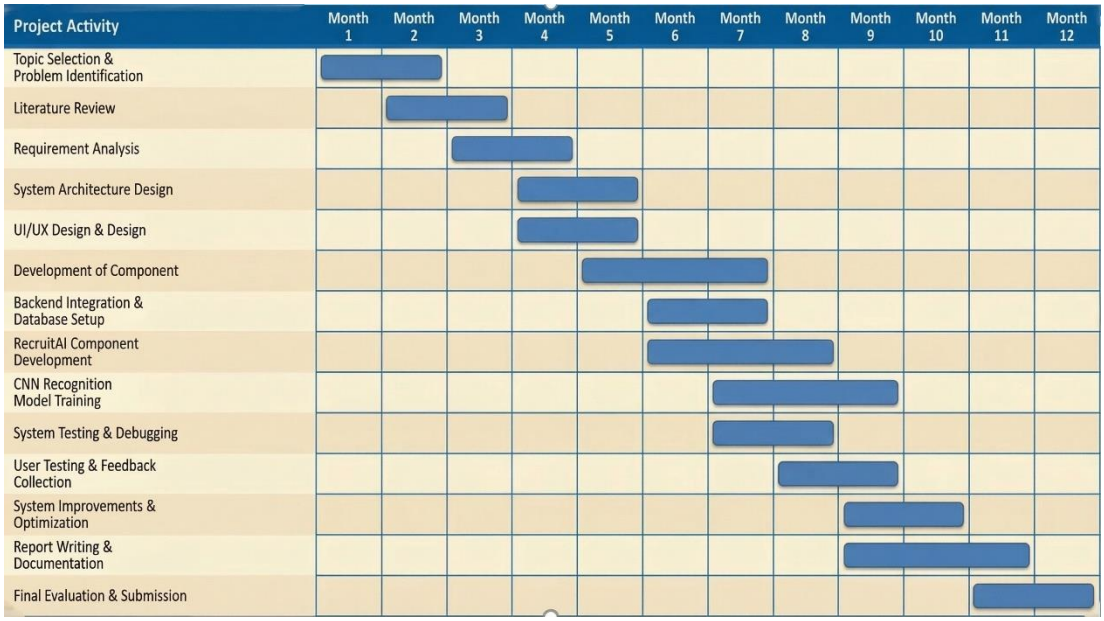


Figure 3. 3. Gantt Chart

4. DESCRIPTION OF PERSONAL AND FACILITIES

4.1. Personnel

- Supervisor and co-supervisor to guide research design, system scope, and evaluation methods.
- Volunteer undergraduate participants to take part in mock interview sessions.
- Optional industry reviewer or lecturer to provide human evaluation of interview performance.
- Research student as the main developer responsible for frontend, backend, LLM integration, and dashboard analytics.

4.2. Facilities

- Laptop or desktop computer with webcam support.
- Stable internet connection for Gemini API access.
- Development tools such as React, Python, FastAPI/Flask, MediaPipe, FER-based libraries, and database tools.
- Secure local or cloud storage for session logs and consented validation data.
- Testing environment for mock interview sessions and dashboard evaluation.

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